Auditable Exact Lower Bounds for Trajectory QCQPs via a Rational Riccati Sweep

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Abstract

Trajectory optimization for spacecraft rendezvous under nonconvex keep-out (*state*) constraints yields nonconvex quadratically constrained quadratic programs (QCQPs) whose global optimality is, in practice, certified only numerically, up to a solver tolerance; we treat the *penalty-coupled* (soft-dynamics) subclass, in which the stage-to-stage dynamics enter the cost rather than as equality constraints and the decision variables are the stage states alone, with no control variable. For this class, the S-procedure positive-semidefiniteness (PSD) test restricted to the problem's band *is* the discrete LQ Riccati (block-Schur) recursion, so running it in exact rational arithmetic gives a tolerance-free certified lower bound in $O(Nd^3)$ exact rational operations; the exact band verdict is proved (the identity itself is classical), while the border-carrying extension that delivers the scalar corner has its soundness asserted and test-pinned, its count read off the sweep's loop bounds. The lower bound is the general-case deliverable: it needs no rationality of the optimum, and on generic keep-out geometry the optimum is irrational, so nothing stronger is available there. On *constructed* instances carrying a designed rational optimum, an aggregate-PSD condition closes the bracket exactly, upgrading the bound to a global-optimality certificate; that result, and the solver-free diagonal-dominance entry to its key hypothesis, are appendix material exercised as exact test oracles. The construction places the emitted certificate $(\lambda^\star, x^\star)$ at per-stage design height L before any datum is derived from it, so the emitted pair has per-entry bit-size $O(L)$ independent of the horizon; the theorem's content is that this designed pair is the certified global optimum and that the verifier recovers it at that size. The certified value the verifier recomputes is not bounded by L, though: it is governed instead by the instance's assembled height together with the height of the placed coordinates and a $\log N$ term, with neither ordering between the assembled and design heights forced. No unconditional bound exists for quadratic optimization at large, which admits 2^n-bit exact optima (whether such instances live in our class is open), so the guarantee is stated conditionally. Aggregate-PSD is implied by a solver-free weak-diagonal-dominance rule, and a counterexample certified in exact arithmetic shows it is a real assumption. The acceptance predicate is machine-checked in Lean, and the pipeline is demonstrated end-to-end on an inverse-designed Clohessy–Wiltshire-coupled keep-out stress instance, checked throughout by an exact implementation that refuses floats. The protocol is sound but not complete: acceptance is a theorem about the deployed numbers, and refusal asserts nothing. Exactness is claimed for auditability and for deciding the equalities — closure and exact contact — that no positive-width enclosure can establish; it is not claimed as the only route to a rigorous bound, nor as a faster one.

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1 Introduction

Guidance for spacecraft rendezvous, proximity operations, and docking (RPOD) plans a trajectory by solving optimal-control problems that, after discretization, are QCQPs: a quadratic cost (tracking, path smoothness) subject to quadratic keep-out constraints such as obstacles and abort-safety balls. Corridors and other convex tubes carry the opposite sign and are not instances of the class treated here. We treat throughout the *penalty-coupled* (soft-dynamics) subclass, in which the stage-to-stage dynamics enter the cost rather than as equality constraints and the decision variables are the stage states alone, with no control variable; hard equality dynamics are outside the class treated here (§5).

These QCQPs are nonconvex, and standard practice — sequential convex programming, or numerical moment–SOS relaxations [23, 21, 42] — returns a solution whose global optimality is either not certified at all or certified in floating point, up to a tolerance. Lossless convexification [1] does certify exactly and analytically, via the maximum principle, but for nonconvex *control*-set constraints — not the state constraints at issue here.

We ask a different question: can the *certificate* of global optimality of a trajectory QCQP be made *exact*, checked in rational arithmetic with no tolerance, so that acceptance is a theorem about the deployed numbers rather than a numerical near-miss? For a single keep-out the answer is known — the S-lemma is exact and its dual is one PSD test [33, 41], with rational recovery [32, 20]; the open case is the *trajectory*, a chain of coupled primitives, where exact certification might fail (a duality gap) or explode (exponential-bit rationals [6]). The question is about checking, not searching. A proposer supplies the candidate — a numerical solver in general, the inverse design of §4 in our demonstrations — and nothing in the certificate route replaces the search that produced it.

The structural observation. The S-procedure certificate for the chain is one bordered PSD condition $M(\lambda, t) \succeq 0$ (§2). Its exact test is a rational $LDL^\top$ that eliminates within the problem's band and carries the affine column alongside; that test is precisely the discrete Riccati / block-Schur recursion of LQ optimal-control numerics [40, 34], and the recursion's terminal arrival cost delivers the single scalar corner condition. Run in exact $\mathbb{Q}$, acceptance becomes a tolerance-free theorem at the $O(Nd^3)$ operation count of one LQ Riccati sweep, modulo the border-carrying extension of §3.

What exactness buys here, and what it does not. Stated up front, because the distinction governs how every result below should be read. When the active keep-outs guard more than one coordinate, the optimum is generically irrational (Remark 2), and the deliverable there is an exact rational *lower bound*, not closure. A rigorous lower bound is not unique to exact arithmetic: interval methods and verified Cholesky factorizations [19, 35] deliver one at fixed precision. This paper claims neither speed nor a tighter bound. What exact arithmetic uniquely delivers is, first, *decidable equality*: closure $t^\star = J^\star$ and exact contact $f_j(x^\star) = 0$ are $\mathbb{Q}$-decidable, and no positive-width enclosure can certify an equality — it can bound a gap below ε, never establish that it is 0. Second, *auditability*: the certificate is a finite rational object that any independent implementation re-verifies bit-for-bit, provided it meets the three re-verifier conditions of §7 — exact rational arithmetic, unbounded integers, the transmitted data — with no tolerance to agree on and no dependence on platform rounding. The closing certificates of Theorem 1 (Appendix A) exercise the equality deliverable on *constructed* instances whose rational optimum is designed in (Definition 3, Appendix A); §5 states what survives without that design.

Delta over the published technical note. The published Podium technical note [28] establishes the exact S-procedure bracket and the rational $LDL^\top$ PSD test for *single* primitives; the mathematics there is classical and presented as such. This paper's delta is the *trajectory* case:

1. **An auditable bound, cheap to check (§2, §3).** On any instance of the class the deliverable is an exact rational lower bound — weak duality needs no rationality of the optimum — carried by a certificate that is a finite rational object, re-verified bit-for-bit by any implementation meeting the three re-verifier conditions of §7. Checking is cheap because of the band identity: the band-restricted PSD test *is* the LQ Riccati recursion, so testing $H(\lambda) \succeq 0$ costs $O(Nd^3)$ exact rational operations (Lemma 1, *proved*); full acceptance of $M(\lambda, t) \succeq 0$ extends this to the bordered sweep at the same $O(Nd^3)$ order, its soundness asserted and test-pinned rather than proved (§3), and the checker itself is sign and zero tests. The protocol is sound but not complete (refusal asserts nothing; Definition 1).

2. **A machine-checked predicate, a shipped checker (§3, §4).** The *acceptance predicate* — a nonnegative-pivot factorization exists iff the matrix is PSD — is *machine-checked* in Lean in both directions; that the shipped sweep computes such a factorization is not machine-checked (§3, [29]). The bordered sweep ships as a trusted checker in the Podium library, and the pipeline is *demonstrated* end-to-end on an inverse-designed CW-coupled stress instance (§4), every reported number reproduced by an executable exact-arithmetic receipt [29].

2 Problem and certificate protocol

Let $x = (x_1, \dots, x_N)$ with each $x_k \in \mathbb{R}^d$ and total dimension $n = Nd$; the optimization is over $\mathbb{R}$, while the problem *data* and the *certificate* are rational. Consider

$$J^\star = \min_x \; f_0(x) \quad \text{s.t.} \quad f_j(x) \ge 0, \; j = 1, \dots, m, \tag{1}$$

with a block-tridiagonal objective coupling only neighbours,

$$f_0(x) = \sum_{k=1}^{N-1} \|x_{k+1} - A_k x_k\|_{R_k}^2 + \sum_{k=1}^{N} (x_k - g_k)^\top W_k (x_k - g_k),$$

with $R_k \succ 0$ the stage-cost weight and W_k the goal weight, and per-stage keep-outs

$$f_j(x) = \|\Pi_j x_{\sigma(j)} - c_j\|^2 - \rho_j^2, \qquad \rho_j > 0, \tag{2}$$

each involving only the single stage $\sigma(j) \in \{1, \dots, N\}$; several keep-outs may guard the same stage. Here Π_j is a fixed coordinate projection selecting the *guarded* sub-block of the stage state: $\Pi_j = I$ is the full-stage keep-out ball, while a guidance keep-out typically guards the position sub-block of a position–velocity state (§4). The neighbour-only coupling confines every nonzero entry to a scalar band of half-width $b = 2d - 1$, and that is the band the sweep of §3 uses.

The certificate's shape can be read off the one-dimensional picture first. At the minimum of a parabola the tangent is horizontal: the derivative vanishes at the minimizer, and that first-order identity is an algebraic fact about the minimizer's own coordinates — a claimed minimizer is checked by evaluating a derivative, not by rerunning any solver. When the minimum sits on the boundary of the feasible set — here, on an active keep-out sphere — the gradient need not vanish. Under a qualification on the active gradients (linear independence of the active constraint gradients, LICQ — the hypothesis of Theorem 1, Appendix A), it is balanced by the active constraints' gradients with nonnegative weights, and those weights are the multipliers λ of the KKT stationarity condition. The balance alone does not certify. With nonconvex keep-outs many points satisfy it: local minima, saddle points, contacts on the wrong branch of a sphere. And which keep-outs are active at the optimum is a combinatorial unknown. The S-procedure certificate (λ, t) defined next is the object that closes the gap: acceptance upgrades "these coordinates balance" to the global statement $t \le J^\star$, and closure of the bracket (Definition 1) upgrades it again, to global optimality of the claimed point — checked in exact arithmetic throughout.

Write $f_0(x) = x^\top P_0 x + q_0^\top x + r_0$ with P_0 block-tridiagonal, and $f_j(x) = x^\top P_j x + q_j^\top x + r_j$. Then $P_j = \Pi_j^\top \Pi_j \succeq 0$ is supported on the single block $\sigma(j)$, and since Π_j is a coordinate projection, $\|\nabla f_j\| = 2\rho_j$ on the contact sphere. These are the structural properties of the keep-outs that the results below use; Lemma 2's (Appendix B) sharp form additionally takes $A_k = I$, $R_k = I$, $W_k = w_k I$ and $\Pi_j = I$. A multiplier vector $\lambda \in \mathbb{Q}^m_{\ge 0}$ and a value $t \in \mathbb{Q}$ certify the lower bound $t \le J^\star$ when

$$M(\lambda, t) = \begin{bmatrix} H(\lambda) & \frac{1}{2}(q_0 - \sum_j \lambda_j q_j) \\ \frac{1}{2}(q_0 - \sum_j \lambda_j q_j)^\top & r_0 - \sum_j \lambda_j r_j - t \end{bmatrix} \succeq 0, \qquad H(\lambda) := P_0 - \textstyle\sum_j \lambda_j P_j,$$

by weak duality: $M \succeq 0$ makes $f_0 - \sum_j \lambda_j f_j - t$ globally nonnegative, so $t + \sum_j \lambda_j f_j$ is a global minorant of f_0. The converse fails: Example 1 (Appendix B) exhibits true lower bounds admitting no level-one certificate. Acceptance of a certificate (Definition 1) is therefore the sufficient direction.

By a generalized Schur complement, $M(\lambda, t) \succeq 0$ holds iff three conditions do:

(i) $H(\lambda) \succeq 0$;

(ii) the linear part $\frac{1}{2}(q_0 - \sum_j \lambda_j q_j)$ lies in range$(H(\lambda))$ — automatic when H is nonsingular, and when H is singular the sweep detects the condition as a consistent zero-pivot row;

(iii) the scalar corner condition $t \le g(\lambda)$, where $g(\lambda)$ is the Lagrangian dual function value at λ.

Remark 1 (Curvature convention)*.* Block tridiagonality of $H(\lambda)$ is forced by the class rather than assumed: every summand of f_0 touches only $\{x_k, x_{k+1}\}$ or x_k alone, each P_j sits in the single block $\sigma(j)$, and block-tridiagonal matrices form a linear subspace. The structure is load-bearing: the band sweep cannot see out-of-band entries, and it returns a wrong verdict on a symmetric matrix that lacks the structure. Under the convention $f_0 = x^\top P_0 x + q_0^\top x + r_0$ used here, with no $\frac{1}{2}$ factor, $H(\lambda)$ is half the Lagrangian Hessian: $\nabla^2_{xx}(f_0 - \sum_j \lambda_j f_j) = 2H(\lambda)$. We therefore call $H(\lambda)$ the *Lagrangian curvature matrix.* The factor is fixed by that convention and is load-bearing wherever H enters quantitatively: H, and not $2H$, is the curvature block of $M(\lambda, t)$, and the pivot bit-sizes §4 measures are those of H. What is invariant under a positive rescaling are the three verdict-level properties invoked below — semidefiniteness, weak diagonal dominance, and pivot signs — since the band pivots are homogeneous of degree one in H.

Definition 1 (Certificate protocol)**.** The *S-procedure certificate* is the pair (λ, t); its acceptance — $\lambda \ge 0$ and $M(\lambda, t) \succeq 0$, checked in exact arithmetic — proves $t \le J^\star$. The *emitted certificate* is the object a prover transmits: the pair $(\lambda, \bar{x})$ of multipliers and a witness. The verifier checks $\lambda \ge 0$ and $f_j(\bar{x}) \ge 0$ for every j. It then recomputes both bounds: a candidate t as the Lagrangian value at $\bar{x}$, which is the route the shipped drivers take, with acceptance re-verifying $M(\lambda, t) \succeq 0$ however t was formed; and the upper bound as $f_0(\bar{x})$. The certified value is therefore never transmitted or trusted. The two outcomes are asymmetric, in the standard sense that the protocol is *sound* but not complete. Acceptance carries the bound as a theorem: no tolerance exists through which a false certificate could pass. A candidate failing any check is *refused*, and a refusal asserts nothing — in particular, it does not prove the candidate suboptimal (Appendix B exhibits an instance whose true optimum admits no level-one certificate (level one: the constant-multiplier S-procedure relaxation used throughout; level two adds multipliers on products of constraints) at all). Throughout, the verifier works in exact rational arithmetic, evaluating f_0 and the f_j and assembling $M(\lambda, t)$ from its own copy of the problem data: a certificate is meaningful only relative to agreed data. The interval $[t, f_0(\bar{x})]$ *brackets* $J^\star$, and the bracket *closes* when $t = f_0(\bar{x})$. Every size or cost claim below names which object it concerns; for claims about a certificate, that is the S-procedure certificate or the emitted certificate.

Standing assumptions. Two assumptions, A1 and A2, are referenced throughout.

Assumption 1 (A1: aggregate-PSD exactness)**.** At a KKT point $x^\star$ with multipliers $\lambda^\star \geq 0$, the Lagrangian curvature matrix satisfies $H(\lambda^\star) \succeq 0$.

Assumption 2 (A2: rational recoverable optimum)**.** $x^\star$ is rational: the active keep-out surfaces $\|\Pi_j x_{\sigma(j)} - c_j\| = \rho_j$ meet the stationarity variety at a rational point.

A1 is the S-procedure exactness condition. Under it, global optimality of $x^\star$ becomes a *conclusion* (Theorem 1), not a hypothesis, and checking A1 is a solver-free PSD test on $H(\lambda^\star)$ that needs no prior knowledge of optimality; for the chain it is implied by the weak-diagonal-dominance rule of Appendix B. A2 is the arithmetic condition that exact closure additionally requires; Remark 2 below shows it fails generically and must be designed in.

Remark 2 (Rational contact is designed, not generic)*.* A2 is arithmetic-geometric and is arranged by construction rather than assumed generically: when the active keep-out guards a sub-block of dimension greater than one, the stationarity variety meets that active sphere at a generically irrational point, so rational contact must be designed in. Designing it in requires $\rho_j \in \mathbb{Q}$, since the standard constructions used in Definition 3(i) (Appendix A) — Pythagorean triples for $d = 2$, stereographic projection of rational parameters for general d — reach rational points only when ρ_j is rational. At guard dimension 2 or 3 a rational non-square ρ_j^2 can leave the sphere with no rational point to design in at all: $x^2 + y^2 = 3$ and $x^2 + y^2 + z^2 = 7$ have none. Dimension 4 or more always admits one, by Lagrange's four-square theorem. A keep-out guarding a single coordinate with rational radius $\rho_j \in \mathbb{Q}$ is the exception: contact there pins that coordinate to the rational value $c_j \pm \rho_j$ directly. What (2)'s data fix is only $\rho_j^2 \in \mathbb{Q}$: with ρ_j^2 a rational non-square, single-coordinate contact is always irrational, while a higher-dimensional sphere may still carry a rational point the constructions just named do not reach, as $(1, 1)$ does on $x^2 + y^2 = 2$. Tolerance-free closure is thus a property of instances whose optimum is rational; §5 states what survives without A2.

2.1 Size measures

All data are rational, and every bit bound in this paper is stated against a height — in the sense of Definition 2 — of a tuple that the bound names. Two statements stand outside the scheme: the per-entry bit-size of the rationalized Clohessy–Wiltshire (CW) state-transition matrix in §4, stated against the denominator cap that produces it, and the imported 2^n-bit lower bound of [6], stated against the problem dimension.

Definition 2 (Height and bit-size)**.**

(a) For a finite tuple T of rationals, its *height* $\mathrm{ht}(T)$ is the least h such that T can be written over one common denominator, $T = (a_1/D, \ldots, a_r/D)$ with D a positive integer and the a_i integers, subject to $\lceil \log_2(D+1) \rceil \leq h$ and $\lceil \log_2(|a_i|+1) \rceil \leq h$. That least h is attained at the least common multiple of the entries' reduced denominators, which is the D the receipts compute. Clearing that denominator turns the tuple into integers of at most h bits — which is what Hadamard's and Edmonds' bounds require.

(b) A single rational p/q in lowest terms has bit-size $\mathrm{bit}(p/q) = \lceil \log_2(|p|+1) \rceil + \lceil \log_2(q+1) \rceil \leq 2\,\mathrm{ht}(p/q)$; that per-entry measure is what Table 2 reports for the emitted certificate.

(c) For a positive *integer* D we write $\mathrm{bitlen}(D) = \lceil \log_2(D+1) \rceil$ for its bit-length, the measure the receipts apply to denominators; read as a rational, $\mathrm{bit}(D) = \mathrm{bitlen}(D) + 1$.

Table 1 collects the named heights the bounds below are stated against; each is the height, in the sense of Definition 2, of a different tuple, and §4 measures all four.

Table 1: The named heights.

Height	tuple measured	where used
L (*per-stage design height*)	the design tuple Definition 3 places on a stage: that stage's placed coordinates, its active keep-out centres c_j, and its active multipliers $\lambda_j^\star$	Theorem 1's bit-size leg, stated against this tuple and no larger one
L_{asm} (assembled height)	the assembled coefficients P_0, q_0, r_0, and the goals g_k they are recovered from	Remark 5's (Appendix A) bounds on the certified value
L_M (bordered height)	the bordered matrix $M(\lambda, t)$ that the sweep eliminates: $L_M = \mathrm{ht}(M(\lambda, t))$	Remark 4's pivot bit-size bound
$\mathrm{ht}(x^\star)$	the placed coordinates taken over *all* stages at once	Remark 5's sharper first inequality

These are different tuples and in general different numbers, and no ordering among L, L_{asm}, L_M and $\mathrm{ht}(x^\star)$ is forced, so each bound below names which height it uses. The corner $r_0 - \sum_j \lambda_j r_j - t$ of the bordered matrix can cancel r_0's denominator against t's, and on the demonstration family of §4 it does. Nor is $\mathrm{ht}(x^\star)$ a renaming of L: L bounds a stage's design tuple one stage at a time, while $\mathrm{ht}(x^\star)$ is taken over the placed coordinates of all stages at once.

3 The certificate is a Riccati sweep

$H = H(\lambda)$ is symmetric block-tridiagonal, with $d \times d$ blocks H_{kk} and couplings $H_{k,k+1} = -A_k^\top R_k$. Its scalar bandwidth is at most $b = 2d - 1$. Equality holds whenever some coupling block has a nonzero $(1, d)$ entry, as it does on the family of §4; structured couplings such as $A_k = R_k = I$ instead sit at half-width exactly d, strictly inside the band whenever $d \geq 2$. The sweep uses the conservative b in either case. Eliminating blocks in increasing order produces the pivots

$$S_1 = H_{11}, \qquad S_k = H_{kk} - H_{k,k-1}\, S_{k-1}^{-1}\, H_{k-1,k} \quad (\text{while } S_{k-1} \succ 0), \tag{3}$$

and, in the positive-definite interior, $H \succ 0$ iff every $S_k \succ 0$; the general semidefinite verdict, including singular boundary pivots, is delivered by the scalar band-$LDL^\top$ of Lemma 1 with its zero-pivot handling.

Equation (3) is the discrete *Riccati / block-Schur* recursion of an LQ problem: S_k is the accumulated (arrival-cost / information-form) curvature matrix after eliminating $x_1, \ldots, x_{k-1}$ — a Schur complement in H, so it inherits the same no-$\frac{1}{2}$ convention as $H(\lambda)$ (Remark 1). That the block-banded KKT factorization equals the Riccati recursion is classical [40, 34]; here it is run in exact $\mathbb{Q}$.

The full certificate is $M(\lambda, t) \succeq 0$, that is, the three conditions of §2. Carrying the affine term $\frac{1}{2}(q_0 - \sum_j \lambda_j q_j)$ through the same sweep makes the terminal corner pivot $g(\lambda) - t$; equivalently, run the sweep at $t = 0$ and read off the arrival cost $g(\lambda)$. All three conditions are thus certified in one tolerance-free rational sweep of $O(Nd^3)$ operations. That sweep is an extension of Lemma 1; its *soundness* is asserted and test-pinned rather than proved, and its operation count is read off the sweep's loop bounds (Remark 3).

Lemma 1 (Band-$LDL^\top$ soundness)**.** *For symmetric H with bandwidth b, the $LDL^\top$ that eliminates only within the band yields the same pivots — hence the same PSD verdict — as the dense test, in $O(nb^2) = O(Nd^3)$ exact rational operations, with exact zero-pivot handling.*

Band preservation is proved below; the content of the lemma is the exact-rational, zero-pivot-handling PSD *verdict*.

Proof. Eliminating pivot k updates $a_{ij} \mathrel{-}= (a_{ik}/a_{kk})a_{kj}$ for $i, j > k$. With bandwidth b, $a_{ik} \neq 0 \Rightarrow i \leq k+b$ and $a_{kj} \neq 0 \Rightarrow j \leq k+b$, so only entries with $i, j \in (k, k+b]$ change and the bandwidth is preserved; by induction all nonzero updates lie in the band, so eliminating only there is exact. At a zero pivot, positive semidefiniteness requires $a_{kj} = 0$ for all $j > k$; bandwidth makes this automatic for $j > k+b$, so checking $j \in (k, k+b]$ suffices. Each pivot touches an $\leq b \times b$ window, giving $O(nb^2)$. □

Remark 3 (The bordered extension)*.* $M(\lambda, t)$ has no bandwidth: its last row and column carry the generally dense $\frac{1}{2}(q_0 - \sum_j \lambda_j q_j)$, so Lemma 1 applied to M gives $O(n^3)$, not $O(Nd^3)$. The bordered sweep instead carries the border as a *separate* column, at one rank-one corner update per pivot. That extension is *asserted here, not proved*. Its soundness is pinned by test in two ways: at the verdict level against the dense M-test, and pivot by pivot against the dense $LDL^\top$ of M at the shortest horizons of §4's family. Its $O(Nd^3)$ count is read off the shipped sweep's loop bounds — each of the n pivots touches a $\leq b \times b$ band window plus $\leq b$ border-column entries and one corner update. No test measures the bordered sweep's operation count; it is the band sweep's count that is measured against its loop bound. The companion note records the extension's status and catalogues the pinning receipts [29].

Both tests ship as trusted checkers in the Podium library. The checkers work in exact rational arithmetic, refuse floats, and reject malformed input — non-rational entries, mismatched shapes, asymmetric blocks — before any verdict. One entry point is the band Hessian test. The other is the corner-carrying bordered sweep: one band $LDL^\top$ carrying the affine column, with the singular-H range condition detected as a consistent zero-pivot row, delivering the full $M(\lambda, t) \succeq 0$ verdict in $O(Nd^3)$ exact rational operations. Both entry points materialize the $n \times n$ matrix before eliminating, so their memory — and hence their total work — is quadratic in n; the $O(Nd^3)$ of Lemma 1 counts the sweep's rational *operations*, which the assembly does not change. A fixed-seed test suite pins both checkers against their dense counterparts — the band-H sweep on the assembled H, the bordered sweep on the assembled M, the latter being the dense M-test the library's reference checker performs — over randomized instances plus dedicated singular-H and zero-pivot cases. The companion note names the checkers, the pinning suite, and the library commit that carries them [29].

Remark 4 (Coarse effectiveness). The sweep's pivots are ratios of consecutive leading minors of the bordered matrix; with zero pivots skipped, the k-th pivot is a ratio of two minors on the retained index set, so the identity survives the zero-pivot branch. Clearing the common denominator that $L_M = \mathrm{ht}\big(M(\lambda, t)\big)$ supplies, their bit-sizes are

$$O\big((n{+}1)(L_M + \log(n{+}1))\big)$$

by Hadamard's inequality and the classical exact-elimination argument of Edmonds [14], and exact verification runs in time polynomial in n and L_M. The order $n{+}1$ is the worst case over the sweep, attained at the *bordered* $(n{+}1) \times (n{+}1)$ matrix itself: only the terminal pivot's minors reach the border, the interior pivots' minors being leading minors of the H-block alone, so bounding every pivot at the bordered order and the bordered height is conservative. The implied constant is not 1: a pivot is a ratio of two minors, so after clearing, the instantiation of the display is about twice the product. The height that governs here is the bordered matrix's, not a per-stage height, and the difference is not cosmetic. Assembly multiplies denominators across the coupling, so on the demonstration family of §4 the bordered height dwarfs the per-stage design height, and no per-stage height bounds even the *entries* of the matrix the sweep eliminates, let alone its pivots. §4 measures those pivots inside the displayed expression formed with L_M at every horizon, and outside the one formed with the design height L. Sharp growth laws are model-sensitive; §4 reports measured growth.

Remark 4's displayed expression takes both the bordered height L_M and the order $n{+}1$ of the bordered matrix because what the sweep costs is bounded through the two together, and through neither alone. This is witnessed by a pair of admissible families (Definition 3, Appendix A), distinct from Table 2's rows, each closing and each constructed by an executable receipt catalogued in the companion note [29]. The first family has bordered matrix entries all in $\{-1, 0, 1, 2, 3\}$, $x^\star = 0$, unit keep-outs contacted at $c_k = -1$, and multipliers alternating 1 and 0. It holds $L_M = 2$ at every horizon while its largest interior pivot grows past any fixed budget measured — 12 bits at $n{+}1 = 9$ and 483 at $n{+}1 = 257$ — so no function of L_M alone bounds the cost. The second family fixes the order at $n{+}1 = 9$ while its multipliers' denominators grow: that raises L_M from 10 to 262 bits and the largest interior pivot from 85 to 4116 with it, so no function of the order alone bounds the cost either.

The acceptance predicate — a factorization with nonnegative pivots exists iff the matrix is PSD — is machine-checked in Lean in both directions; that the shipped sweep computes such a factorization is not, and the companion note records what is machine-checked and under which hypotheses [29].

4 Pipeline validation: an inverse-designed CW-coupled instance

We instantiate (1) as a *certification stress instance with Clohessy–Wiltshire structure* — not a flyable maneuver. Consecutive stages are coupled through the planar CW state-transition matrix Φ (mean motion $n_0 = 10^{-3}$ rad/s and sampling phase $\tau = n_0\,\Delta t = 1/2$, so $\Delta t = 500\,\mathrm{s}$) via the soft-dynamics *full-state penalty* $\|x_{k+1} - \Phi x_k\|_R^2$: there is no control variable and no thrust model, the keep-out radius cycles $\{1/2, 1, 3/2\}$ per stage, and the instance is *constructed by inverse design* (each stage's guarded contact placed on a rational keep-out-sphere point, goals recovered from stationarity, multipliers chosen so A1 holds — a *benign* instance in Appendix B's vocabulary). The stage state is $x_k = (x, y, v_x, v_y) \in \mathbb{Q}^4$ and the keep-out guards its *position* sub-block: Π_k in (2) selects (x, y), giving $\|\Pi_k x_k\|^2 - \rho_k^2 \geq 0$ about the target at the LVLH origin, with the two velocity

Table 2: Exact-rational trajectory certificates (all verdicts exact over $\mathbb{Q}$). "emitted bits" is the worst-entry bit-size of the emitted $(\lambda^\star, x^\star)$; "value bits" is the size of the certified value $t^\star$ the verifier recomputes (Remark 5). The size column carries both units: N is the number of stages, d the per-stage state dimension, and $n = Nd$ the number of scalar variables. The regime column places each family in the vocabulary of Appendix B: *benign* means A1 holds at the optimum, *A1 violated* means it fails and the checker refuses. Rows 2 and 3 are constructed rational-sphere families (Definition 3); row 1 is not — the scalar shared-state family leaves inactive stages with solved coordinates. It is benign, and covered by Theorem 1's closure leg, but not by its $O(L)$ bit-size bound, which is stated only for constructed families; so its two bit columns are empirical measurements. The CW row's ranges are over all 23 horizons $N = 2, \ldots, 24$; rows 1 and 2 are over the horizons the measurement receipt samples, up to the max size shown.

Family	regime	max size	closes	emitted bits	value bits
Scalar shared-state	benign	N=129 (n=129)	yes	13–18	13–26
Vector shared-state ($d = 1..4$)	benign	n=Nd=96	yes	3–4	15–32
CW-coupled keep-out	benign	N=24 (n=96)	yes	6–8	931–1089
Aggregate-PSD violated	A1 violated	—	no (refuses)	—	—

coordinates placed rationally by the design. It demonstrates the certification pipeline on a rational-optimum instance whose *coupling matrix* is a genuine CW state-transition matrix; the *trajectory* is not CW-consistent and is not meant to be.

A stage is one sampled state: the horizon variable N counts state samples taken $\Delta t = 500\,\mathrm{s}$ apart, and one sampling step advances the reference orbit's phase by $\tau = 1/2\,\mathrm{rad}$, so about 12.6 steps span one orbit. The N stage states are joined by $N-1$ coupling residuals, so a horizon spans $(N-1)\,\Delta t$. At the largest demonstrated horizon, $N = 24$, that is 11500 s: about 3.2 h, or 1.8 orbits in the model's units. The same horizon carries $n = Nd = 96$ scalar variables at $d = 4$ per stage. The translation to time is specific to the CW-coupled family: the shared-state families of Table 2 are abstract chains with no sampling interval.

The design places the optimum where stationarity requires it: at N=24 the per-step dynamics residual $\|x^\star_{k+1} - \Phi x^\star_k\|$ runs 134.8–244.1 against states of norm 0.559–1.572, and the goals recovered from stationarity reach $\|g_k\| \approx 1.01 \times 10^5$. Units are nominal — positions in metres, velocities in metres per second — so $R = I_4$ and $W_k = w_k I_4$ are not dimensionally homogeneous weights and the norms just quoted mix the two, consistent with the instance not being a flyable maneuver. So this does not discover an optimum for an operational CW problem, and the goal term is a goal-attraction penalty, not a tracking objective. The CW state-transition matrix is rationalized entrywise to denominators at most 10^5, giving per-entry bit-size at most 42. The rationalization breaks the symplectic structure: after rounding, $\det\Phi = 1 + 1.2 \times 10^{-8}$ exactly, against $\det\Phi = 1$ for the CW flow. The certificate is therefore a theorem about the *rational model only*: it certifies optimality and keep-out satisfaction for the rationalized dynamics, not for the continuous CW flow. The rounding moves each entry by at most 2.8×10^{-10} in absolute value and 2.0×10^{-7} relative; it is the 10^5 denominator cap, not the 42-bit entry size, that sets those figures.

Everything is checked in $\mathbb{Q}$ by the three routes of §3 — the band-H test, the corner-carrying bordered sweep on the full M, and the dense M-test — which agree on every row of Table 2. At every one of the 23 horizons $N = 2, \ldots, 24$ the bracket closes, so weak duality already forces $g(\lambda^\star) = t^\star$ and a terminal corner pivot of exactly 0; as a sanity check the bordered sweep refuses $t^\star + 10^{-6}$ at every horizon, and the guarded position sub-blocks of the optima lie exactly on the keep-out spheres.

The emitted pair is small in every family — 18 bits or fewer — while the recomputed CW *value* ($\sim 10^3$ bits) comes from the inverse design, not from the raw data. The goal term $r_0 = \sum_k w_k \|g_k\|^2$ *squares* the denominators recovered from stationarity: $\mathrm{den}(t^\star)$ and $\mathrm{den}(r_0)$ coincide at every horizon $N \geq 6$ (both 519 bits at N=24), and $\mathrm{den}(t^\star)$ divides $\mathrm{den}(r_0)$ at 22 of the 23 horizons — the exception is N=5, where the two differ by a factor 6 the other way. Per Remark 5 it is the *denominator* that is bounded in N: $\mathrm{den}(t^\star)$ stays inside 450–526 bits at all 23 receipt-covered horizons. The bit-size keeps the remark's residual $\log N$ drift, measured at 931–1089 bits over all 23 demonstration horizons. The verifier's multiplier reconstruction (Theorem 1) shows the same split between the design and assembled heights. The solve's *operands* stay at the design height: 5–6 bits against $L = 6$ on this family, and 259 bits against $L = 258$ on the inverted family noted inside Definition 3's closing paragraph. They stay there because stationarity makes the objective gradient the multiplier combination of the active keep-out gradients, which are design-tuple data. The *formation* of that gradient from the assembled coefficients carries the assembled height instead: on this family its two terms run 242–276 bits against $L = 6$ and cancel to the design height, while on the inverted family they do not cancel, the formed gradient staying at 259 bits against $L_{\mathrm{asm}} = 3$–8.

The heights, measured. The four separate heights of §2.1 are "in general different numbers" on this family in the strongest sense: at every one of the 23 horizons its four tuples carry four pairwise distinct heights, in the strict order $\mathrm{ht}(x^\star) < L < L_M < L_{\mathrm{asm}}$; Table 3 reports the measured ranges.

Table 3: The four heights of §2.1, measured on the CW-coupled demonstration family over all 23 horizons $N = 2, \ldots, 24$; they are pairwise distinct at every horizon, in the strict order $\mathrm{ht}(x^\star) < L < L_M < L_{\mathrm{asm}}$.

Height	measured	what it governs here
$\mathrm{ht}(x^\star)$	5 bits at every horizon	Remark 5's sharper first inequality
L	6 bits, attained at every horizon (per-stage values 3, 5 or 6)	Theorem 1's $O(L)$ leg; Table 2's emitted 6–8 bits
L_M	277–284 bits	Remark 4's pivot expression
L_{asm}	483–563 bits	Remark 5's bounds on the certified value; carries the 519-bit $\mathrm{den}(r_0)$

The per-stage *design* tuple of Definition 3(iii) — a stage's placed coordinates, its keep-out centre (the LVLH origin here) and its multiplier — has height at most $L = 6$ bits at every stage of every horizon, the maximum attained at every one of them. Table 2's emitted 6–8 bits sit next to that figure by construction rather than by measurement: the emitted pair is part of the very tuple L is taken over, so $\mathrm{bit}(\cdot) \leq 2L$ (Definition 2) holds identically, here and on any constructed family. What the horizons do show is the point actually at issue: neither figure grows with N. The assembled coefficients (P_0, q_0, r_0, g) have height $L_{\mathrm{asm}} = 483$–563 bits, consistent with Definition 3(iv), which makes r_0 derived data of the assembled height rather than of L: the 519-bit $\mathrm{den}(r_0)$ above is carried by L_{asm} and could be carried by neither the design height nor L_M. And the assembled bordered matrix $M(\lambda^\star, t^\star)$ that the sweep eliminates has height $L_M = 277$–284 bits, the smaller of the two assembled figures because its corner $r_0 - \sum_j \lambda_j^\star r_j - t^\star$ cancels those two large denominators against each other.

Instantiating Remark 4's expression at the bordered order $n{+}1$ gives $(n{+}1)\big(L_M+\lceil\log_2(n{+}1)\rceil\big)$, running 2529–28227 bits over the horizons. The measured pivots run from 777 bits at $N{=}2$ to 17893 at $N{=}24$ and stay inside that expression at all 23 horizons. Because the bracket closes, the terminal corner pivot $g(\lambda^\star)-t^\star$ is exactly 0 at every horizon, so the figure reported is the largest *interior* pivot of the corner-carrying sweep. The same expression formed with the design height L runs only 90–1261 bits and is exceeded at all 23 horizons. The pair of measurements $L_M = 277$–284 bits against $L = 6$ already says that on this family no per-stage height bounds even the *entries* of the bordered matrix, let alone its pivots. Here it is the bordered height, and not a per-stage height, that Remark 4's expression must be instantiated with.

That ordering is a property of the CW assembly and not of Definition 3, which forces neither direction: an exact receipt exhibits an admissible family, closing on all three checkers, on which $L_M \leq 8$ bits stands against a design height $L = 257$–258, so there the per-stage design height bounds every entry of the bordered matrix [29]. Remark 5's three inequalities hold at all 23 horizons as well — the compact $\mathrm{bitlen}(\mathrm{den}(t^\star)) \leq L_{\mathrm{asm}} + 2L$, the value bound $\mathrm{bit}(t^\star) \leq 2(L_{\mathrm{asm}} + 2L) + \lceil\log_2 N\rceil + \lceil\log_2(3d^2{+}d{+}1)\rceil$, and the sharper first form $\mathrm{bitlen}(\mathrm{den}(t^\star)) \leq L_{\mathrm{asm}} + 2\,\mathrm{ht}(x^\star)$ — as does the divisibility step behind them. The pivots themselves grow measurably linearly in the horizon — 780 bits at $N{=}2$ to 17889 at $N{=}24$, measured over the blocks S_k of (3) rather than over the scalar pivots, hence a few bits either side of the bordered figures above; this is reported as data.

Reproducibility. Each row of Table 2 is constructed by a named driver in exact rational arithmetic — the three closing rows certified, the refusal row refused — and each driver verifies its emitted certificate through the exact dense M-test. The refusal row's instance takes $\lambda_k > w_k$ at $N{=}4$, $d{=}2$. Measurement receipts recompute both bit columns at the sizes Table 2 headlines, re-verify every closing row through all three routes (the dense reference M-test, the trusted band-H sweep, and the trusted bordered-M sweep), and recompute every denominator statistic quoted in this section at all 23 horizons; a shipped test suite executes each receipt and asserts its published constants. The trusted checkers refuse floats, and the CW state-transition matrix alone is synthesized in floating point and rationalized before any certification, as above. The receipts accompany this record as supplementary material, and the trusted checkers ship in the Podium repository [30], described in [31]. The companion note catalogues every receipt by role and records the bundle's conventions, including the bit-size convention and the library commit that carries the band checkers [29].

5 Scope and limitations

Without A2, the deliverable is a bound, not closure. Without A2, closure is unattainable; the failure of A2 is generic when the active keep-out guards a sub-block of dimension greater than one (Remark 2). The deliverable does not collapse: the sweep still returns an exact, tolerance-free rational *lower bound*, since weak duality needs no rationality of the optimum. A keep-out guarding a single coordinate with rational radius instead pins the active coordinate to a rational value, leaving the remaining coordinates a rational linear solve. Quantifying the resulting bracket (rates in the rationalization budget, behaviour at the relaxation-gap floor) is outside this paper's scope.

Verification bit-complexity is measured and coarsely bounded, not sharply bounded. This paper *measures* verification cost — the linear pivot growth of §4 — and bounds it coarsely (Remark 4); sharp growth laws and their model sensitivity are outside scope. That verification cost can grow along the horizon while the emitted certificate stays bounded is visible in the demonstration of §4.

The demonstration family's mildness is tied to its shared transition matrix. The CW-coupled family couples every stage through one shared rationalized transition matrix, computed for a circular reference orbit. An eccentric-reference variant of the same construction — the Yamanaka–Ankersen transition matrix, a different matrix at every stage — reproduces the CW pivot sequence bit for bit at $e = 0$. At the probed eccentricities $e \in \{0.1, 0.3, 0.5\}$, rationalized to the same realized fidelity, its sweep pivots grow at $\approx$ 1000–1050 bits per stage against the CW family's $\approx$ 778, and the driver is the per-stage time-variance of the transition matrix rather than the eccentricity magnitude. On the CW family the certified value's denominator is horizon-bounded (§4); on the variant the value's bit-size grows at $\approx$ 1000 bits per stage, so the compactness of the recomputed value does not carry over. An executable receipt catalogued in the companion note pins these figures [29].

The per-stage Hadamard/Cramer reconstruction bound is not included. A per-stage Hadamard/Cramer bound on the verifier's multiplier reconstruction is weaker than the $2L$ bound the design order gives, a consequence of the design order rather than a route to it, and not needed for either leg of Theorem 1.

Bienstock realizability inside the class is not established. Quadratic optimization at large admits 2^n-bit optima [6]; whether such a gadget can be realised as an instance of (1) is not settled either way. That is why Theorem 1 is stated conditionally, and why this paper does not claim conditioning is unavoidable.

Hard equality dynamics are outside the class. Hard equality dynamics are outside (1) by definition (Appendix B); readers of the banded-KKT literature will recognise that Wright and Rao–Wright–Rawlings [40, 34] factorize precisely that equality-constrained system, in floating point. The exact-certificate treatment of hard dynamics itself is outside scope.

A1-failing keep-out geometry needs level two. Under A1, multiple keep-outs per stage already close (Theorem 1); when a stage's keep-out geometry itself violates A1 — several exclusion balls on one stage, two quadratic constraints where the S-lemma wants one, the sign-flipped analogue of the two-*inclusion*-constraint CDT subproblem [8, 2] — level-one certification gaps, and an exact level-two treatment is outside scope.

6 Related work and positioning

Exact-rational certificates for optimization. The direct antecedent of tolerance-free certificates for optimization outputs is the exact rational LP/MIP line: VIPR's machine-verifiable exact certificates for integer-programming results [9], exact rational MIP solving in SCIP [15], and the exact LP solver QSopt_ex [4]; the bit-size control of exact elimination goes back to Edmonds [14]. Certificates measured by their verification complexity are the program of [13]. Ours extends the exact-certificate discipline from linear to a structured nonconvex quadratic class, with the banded-KKT/Riccati structure doing the work.

Trajectory SDP relaxations and tightness. Sparse moment–SOS for trajectory optimization [21] and perception [42] is the closest numerical program: floating-point, tolerance-based certificates — STROM reports certified suboptimality below 1% for most of its five trajectory problems, near 1% for the hardest (car back-in), and Yang and Carlone report a sparse relaxation that is *empirically exact*, deeming a rounded solution globally optimal at a relative suboptimality below 10^{-3} — obtained from level-two (sparse second-order) relaxations that close gaps left open at level one, with tightness observed rather than proved. The two lines are complementary: exact-arithmetic acceptance on benign, rational-optimum (constructed) instances versus broad numerical coverage. On the same constraint geometry as Appendix B, Graesdal et al. [16] give necessary *and* sufficient tightness conditions for a *single*-sphere-obstacle trajectory SDP relaxation; tightness in their own many-obstacle experiments is measured empirically, not obtained from that theorem. Ours is only sufficient and holds with any number of keep-outs per stage — of which LICQ leaves at most d active on each — but is solver-free and checkable in exact arithmetic along the sweep, and their single-obstacle conditions delimit what a sharp benign boundary could look like. In the adjacent domain of spacecraft conjunction avoidance, Vega et al. [37] solve collision-avoidance exclusion QCQPs via a Shor relaxation observed empirically tight — on a different formulation, with control variables and conjunction-derived constraints, so not a measurement of A1 on our class. We share no benchmark with any of these, so this paragraph is positioning, not comparison. A general tightness framework for QCQP SDP relaxations [38], graph-structural exactness conditions [36], exactness on forest structures [5], and local-to-global exactness from clique-wise sub-SDPs [22] position A1 within a broader theory (a path is a tree, but our concave keep-outs need not satisfy their hypotheses); chordal conversion and PSD completion [43, 17] address the structural SDP side. Lossless convexification [1] exactly convexifies nonconvex *control*-set constraints via the maximum principle — complementary in mechanism to exact certificates for nonconvex *state* constraints.

Rational recovery and exact SDP. Exact rational SOS/PSD certificate recovery is due to [32, 20], with RealCertify the standard tool [25]: its multivariate route computes an approximate decomposition of a perturbed input with an *arbitrary-precision* SDP solver (SDPA-GMP) and recovers an exact rational one symbolically, while its univariate `univsos1` uses no SDP at all — a recursive construction over $\mathbb{Q}$ from root isolation and quadratic under-approximations. Exact algorithms for LMIs and the SPECTRA library are the exact-SDP lineage [18]. Bit-size bounds for exact SOS certificates — univariate [27], multivariate [26, 11] — do not exploit band structure, an orthogonal axis to Theorem 1; machine-checkable SOS certification is developed in [24], the Coq-interfaced companion to the template method of [3].

Verified solvers and a-posteriori numerics. A complementary line proves the *solver* correct before it runs (credible autocoding and verified convex-optimization codes [39, 12, 10]); Roux et al. [35] validate solver output a posteriori via a floating-point Cholesky with proved rounding-error bounds — rigorous but tolerance-based. The a-priori and a-posteriori guarantees compose with ours rather than compete. On *why exactness*: a rigorous inequality certificate need not be exact. Interval arithmetic or a verified Cholesky [19, 35] certifies $H(\lambda) \succ 0$ with a positive margin, hence a rigorous lower bound, at fixed precision. A positive-margin criterion cannot reach the semidefinite boundary, and the closing certificates of Theorem 1 sit exactly there: at closure the corner Schur complement is exactly 0, and singular H is the case §2 handles through the consistent zero-pivot row. Fixed precision reaches the boundary only in a narrow case: the operations witnessing the boundary contact must be exact in the working precision, so that the boundary quantity is enclosed with zero width. Pivots away from the boundary may round, their positive-width enclosures absorbed by their positive margins. Even in that case a Cholesky does not certify the boundary, because it stops at an exactly computed zero pivot: on the singular H with rows $(1, -1)$ and $(-1, 1)$, where an $LDL^\top$ and the symmetric eigensolver are exact, LAPACK's `dpotrf` declines the factorization, reporting the order-2 leading minor not positive definite. The rationalized CW instance of §4 is not of that boundary-case kind at all: its entry denominators are not powers of two. Exact receipts pin each of these boundary facts; the companion note catalogues them [29]. Exact rational arithmetic is likewise essential for the *equality* deliverables, decidable closure $t^\star = J^\star$ and exact contact $f_j(x^\star) = 0$. Both are $\mathbb{Q}$-decidable and auditable, but no finite-precision enclosure of positive width can establish them: such an enclosure can bound a gap below ε, never certify it is 0. A zero-width enclosure can, and fixed precision delivers one only in the exact-witness case named above. We claim exactness for auditability and decidable equality, not as the only route to a rigorous bound.

7 Conclusion

The exact-rational certification of a *penalty-coupled* block-banded trajectory QCQP is an LQ Riccati sweep run in $\mathbb{Q}$. The sweep is sound at the band-Hessian test (Lemma 1); the border-carrying extension's soundness is asserted and test-pinned (§3); the acceptance predicate is machine-checked [29]; and at a constructed rational optimum the bracket closes under an aggregate-PSD condition implied by a local diagonal-dominance rule (Theorem 1, Lemma 2).

Operationally, exactness buys *auditability*: the certificate is a finite rational object that *any* independent implementation *can* re-verify bit-for-bit, with no tolerance to agree on and no dependence on platform rounding. Three conditions govern that re-verification — exact rational arithmetic, unbounded integers, and the transmitted data; they are the re-verifier's and not the certificate's, and each is load-bearing. The same sweep transcribed operation for operation into IEEE-754 doubles *refuses* a receipt-pinned closing certificate — the disjoint-contact two-keep-out instance of Appendix A — its terminal corner coming out -4.44×10^{-16} rather than exactly 0: a tolerance-free test has no tolerance to absorb its own rounding. The $t^\star$ the verifier recomputes on the demonstration family of §4 runs 931–1089 bits, which no fixed-width arithmetic can hold, while the emitted pair itself stays at 6–8 bits. And Definition 1 conditions the certificate on agreed data, so a re-verifier that re-derives the rationalized Φ of §4 instead of reading the transmitted rationals certifies a different instance. So conditioned, the acceptance of a penalty-coupled trajectory QCQP in the certified class becomes a reproducible theorem about the deployed numbers rather than a solver verdict that must itself be trusted.

Appendix A Exact closure on designed instances

Role in this paper. The closure theorem below is conditional on a rational optimum (A2), and Remark 2 shows A2 fails generically once an active keep-out guards more than one coordinate: the closing instances are designed, their rational optimum placed by construction. Their value is as exact test oracles — instances whose answer is known perfectly, so any deviation by an implementation is a defect; the float-refusal receipts were obtained exactly this way. On generic instances the deliverable remains the exact lower bound of §3.

Definition 3 (Constructed rational-sphere family)**.** A *constructed rational-sphere family* is a family of instances of (1) carrying a distinguished rational point $x^\star$, the *designed point* — Theorem 1, not this definition, is what makes it the optimum — in which the following four conditions hold.

(i) Every keep-out is *active* at $x^\star$, with radius $\rho_j > 0$, and each *guarded* contact point $\Pi_j x^\star_{\sigma(j)}$ is placed at a rational point of its keep-out sphere via a Pythagorean or stereographic parametrization, the remaining stage coordinates being placed rationally as well, so that each stage's placed coordinates have height at most L.

(ii) At every stage the active constraint gradients $\nabla f_j(x^\star)$ are linearly independent. For one keep-out per stage this is automatic, since $\rho_j > 0$; for several keep-outs per stage it is a condition on the contact geometry. Since those gradients are supported on the one block, it caps the number of active keep-outs on a stage at d.

(iii) The multipliers $\lambda_j^\star \geq 0$ are chosen rationally so that, together with the stage's placed coordinates and keep-out centres, they form a per-stage design tuple of height at most L, and so that aggregate-PSD (A1) holds.

(iv) Every W_k is invertible, and the goals g_k are then recovered from stationarity, so the g_k — and with them q_0 and r_0 — are *derived* data, carried by the assembled height L_{asm} rather than by L.

The definition forces no ordering between L and L_{asm}: on the demonstration family (§4) L_{asm} is by far the larger, while an executable receipt exhibits an *inverted* family, admissible under this definition, on which it is the smaller [29]. No coordinate is left inactive to be solved for, so the emitted pair $(\lambda^\star, x^\star)$ is rational, of per-stage height at most L and hence of per-entry bit-size at most $2L$ (Definition 2), by construction.

Theorem 1.

(i) Closure. *Under Assumptions 1 and 2, with linear independence of the active constraint gradients (LICQ) at $x^\star$, the exact-rational bracket closes: there exist rational ($\lambda^\star \geq 0$, $t^\star = J^\star$, $x^\star$) with $M(\lambda^\star, t^\star) \succeq 0$, $f_j(x^\star) \geq 0$ and $t^\star = f_0(x^\star) = J^\star$ — an accepting certificate in the sense of Definition 1 — with any number of keep-outs per stage, of which LICQ leaves at most d active on each.*

(ii) Emitted size and recovery. *For a constructed rational-sphere family (Definition 3) the emitted certificate $(\lambda^\star, x^\star)$ additionally has* per-entry *bit-size $O(L)$, with a constant depending on d but not on N, and a verifier holding the instance data and the transmitted witness $x^\star$ recovers the same $\lambda^\star$ within that bound, so the transmitted $\lambda^\star$ need not be trusted.*

Proof. Closure. By Assumption 1, $x^\star$ is a KKT point with multipliers $\lambda^\star \geq 0$; for a constructed family they are supplied outright, by Definition 3(iii)–(iv). A constraint qualification is not what produces them, since the theorem does not assume $x^\star$ is a local minimizer: optimality is the conclusion. By A1 the Lagrangian $\mathcal{L}(x) = f_0(x) - \sum_j \lambda_j^\star f_j(x)$ is convex and $x^\star$ is a stationary point, hence a global minimiser: $\min_x \mathcal{L} = \mathcal{L}(x^\star) = f_0(x^\star)$, the last step by complementary slackness. Thus $M(\lambda^\star, t^\star) \succeq 0$ at $t^\star = f_0(x^\star)$, and weak duality gives $t^\star \leq J^\star$, while feasibility gives $J^\star \leq f_0(x^\star) = t^\star$. Hence $t^\star = J^\star = f_0(x^\star)$.[1] Nothing in this argument uses one keep-out per stage: with several active keep-outs on a stage the multipliers enter $\mathcal{L}$ additively and complementary slackness holds term by term. Two exact receipts with two active keep-outs on one stage accompany this [29]: one with disjoint-coordinate contacts, closing at $t^\star = 169/320$, and one whose contacts' gradients share their whole support, closing at $t^\star = 5773/3600$.

[1]The mechanized counterparts of the weak-duality step and the attainment leg, and the hypotheses those theorems take, are itemized in the companion note [29].

Rationality of the multipliers. The data and $x^\star$ are rational, so stationarity $\nabla f_0(x^\star) = \sum_j \lambda_j \nabla f_j(x^\star)$ together with complementary slackness ($\lambda_j = 0$ at every inactive j) is a rational linear system in the active multipliers, and under LICQ — item (ii) of Definition 3 for the constructed families — it has a unique solution, which is therefore rational.

Bit-size of the emitted pair. A2 alone does not bound the height: for quadratic optimization at large the KKT system is polynomial, and Bienstock-type gadgets satisfy A2 with 2^n-bit optima [6]. In a constructed family the design order of Definition 3 settles the emitted pair directly: $x^\star$ by (i) and $\lambda^\star$ by (iii) are *placed* rationally at per-stage height at most L, and only then are the goals recovered by (iv). The placement leaves nothing to solve for: every keep-out being active, the guarded coordinates sit on their keep-out sphere and the rest are placed freely, with no inactive coordinate left over. So $(\lambda^\star, x^\star)$ has per-entry bit-size $O(L)$, independent of N.

What the verifier reconstructs. What needs an argument is that $\lambda^\star$ is exactly what a verifier holding the instance data and the transmitted witness $x^\star$ reconstructs. Fix a stage s, let A_s be the keep-outs active there and $m_s = |A_s|$; their gradients $\nabla f_j(x^\star) = 2\Pi_j^\top(\Pi_j x_s^\star - c_j)$ are supported on block s, hence live in $\mathbb{Q}^d$, and have Euclidean norm $2\rho_j > 0$, and by (ii) they are linearly independent, so $m_s \le d$. Write $G_s \in \mathbb{Q}^{d\times m_s}$ for their block-s restrictions and $v_s = [\nabla f_0(x^\star)]_s$. Stationarity reads $G_s \lambda^\star_{A_s} = v_s$ with G_s of full column rank, so the normal equations

$$\left(G_s^\top G_s\right)\lambda^\star_{A_s} = G_s^\top v_s$$

have $\lambda^\star_{A_s}$ as their unique solution, and the verifier reconstructs the emitted multipliers rather than trusting them. What it reconstructs is the *placed* $\lambda^\star$ itself, so Definition 3(iii) together with bit $\le 2\,\mathrm{ht}$ (Definition 2) bounds it by $\mathrm{bit}(\lambda_j^\star) \le 2L$ — the design order, not the solve, is what makes the recovered object small. Nothing here asks the active gradients on a stage to have disjoint supports: the shared-support receipt above exhibits a Definition-3-admissible stage on which they share every coordinate, and checks that the normal-equation solve recovers $\lambda^\star$ there. What that bound does *not* cover is the cost of the same solve run from the assembled instance alone. That solve's *operands* stay at the design height on every family measured, and not by accident: stationarity makes the objective gradient $[\nabla f_0(x^\star)]_s$ the multiplier combination of the active gradients $2\Pi_j^\top(\Pi_j x_s^\star - c_j)$, which are design-tuple data. What carries the assembled height is the *formation*: a verifier holding only the assembled coefficients and the transmitted witness builds $[\nabla f_0(x^\star)]_s$ as $2(P_0 x^\star)_s + q_0^{(s)}$. That formation is bounded through the assembled height together with the height of the placed coordinates and active centres, not through L alone, and which of the two dominates it is a property of the instance; §4 reports the measured figures. Acceptance recomputes $t^\star$ and runs the exact $LDL^\top$; the size of those intermediates is a *verification* cost (Remark 4), not a property of the emitted pair. □

Remark 5 (Size of the certified value). Write $\mathrm{den}(\cdot)$ for the lcm of the denominators of a tuple of rationals. At a *closing* certificate — where complementary slackness makes the Lagrangian value the verifier recomputes equal to $f_0(x^\star)$ — the certified value $t^\star = f_0(x^\star)$ is a sum of N stage terms, the k-th the quadratic form in $x^\star$ carried by the *assembled* stage coefficients $(P_0^{(k)}, q_0^{(k)}, r_0^{(k)})$: block row k of P_0, block k of q_0, and a share $r_0^{(k)}$ of the constant, block-tridiagonality making that row reach only stages $k-1$, k and $k+1$. A merely bracketing t additionally carries $\mathrm{den}(\lambda)$ and the keep-out coefficients' $\mathrm{den}(q_j)$ and $\mathrm{den}(r_j)$, with $q_j = -2\Pi_j^\top c_j$ and $r_j = \|c_j\|^2 - \rho_j^2$ read off (2), so that $\mathrm{den}(r_j) = \mathrm{den}(\rho_j^2)$ for the origin-centred keep-outs of §4 and not in general. Any splitting of the constant with $\sum_k r_0^{(k)} = r_0$ and $\mathrm{den}(r_0^{(k)}) \mid \mathrm{den}(r_0)$ serves — for instance carrying all of r_0 on one stage — and the receipt takes the conservative $\mathrm{den}(r_0)$ at every stage. In four steps:

(1) Each summand's denominator divides $\mathrm{den}(P_0^{(k)}, q_0^{(k)}, r_0^{(k)}) \cdot \mathrm{den}(x^\star)^2$, the form being quadratic in the witness, and a finite sum's denominator divides the lcm of the summands', so

$$\mathrm{den}(t^\star) \;\Big|\; \mathrm{lcm}_{k \le N}\, \mathrm{den}\big(P_0^{(k)}, q_0^{(k)}, r_0^{(k)}\big) \cdot \mathrm{den}(x^\star)^2;$$

hence $\mathrm{den}(t^\star)$ is bounded in N whenever those denominators come from a bounded set, as all families of Table 2 do.

(2) Stated against the heights of §2.1 — the *coefficient* height L_{asm}, not the bordered L_M, since it is the coefficient denominators that appear above — the right-hand side divides the common denominator that L_{asm} supplies times $\mathrm{den}(x^\star)^2$, so $\mathrm{bitlen}(\mathrm{den}(t^\star)) \le L_{\mathrm{asm}} + 2\,\mathrm{ht}(x^\star)$. Here $\mathrm{ht}(x^\star)$ is taken over all stages at once, whereas L bounds a stage's design tuple one stage at a time, so the more compact $\mathrm{bitlen}(\mathrm{den}(t^\star)) \le L_{\mathrm{asm}} + 2L$ additionally asks that the placed coordinates share one denominator along the horizon — as they do on the family of §4.

(3) The numerator adds only the $\log N$ of the sum over stages and the log of the number of monomials one block row contributes — at most $3d^2{+}d{+}1$, since block-tridiagonality leaves block row k at most $3d$ nonzero columns across its d rows, plus q_0's block of d and the constant — giving $\mathrm{bit}(t^\star) \le 2(L_{\mathrm{asm}} + 2L) + \lceil \log_2 N \rceil + \lceil \log_2(3d^2{+}d{+}1) \rceil = O(L_{\mathrm{asm}} + L + \log N)$, so whenever L_{asm} is bounded in N, as it is on the demonstration family of §4, the value's size grows at most logarithmically along the horizon.

(4) The governing quantity is *not* the lcm of the raw data denominators, and cannot be: the assembled coefficients are products and quotients of the data — $A_k^\top R_k A_k$ multiplies two denominators, and recovering g_k from stationarity (Definition 3(iv)) divides by W_k, so a data *numerator* can become a coefficient denominator — after which the goal term squares them. On the CW family of §4 the raw-data lcm is 130 bits while $\mathrm{den}(t^\star)$ is 450–526 bits; the displayed divisibility holds at every one of the 23 horizons $N = 2, \ldots, 24$, and $\mathrm{den}(t^\star)$ divides the raw-data lcm at none of them.

§4 checks all three inequalities, and the divisibility step behind them, at every horizon of the demonstration family.

Why Remark 5 is stated against the heights it names, and why its bit bound carries a horizon term, is settled by admissible closing families, each constructed by an executable receipt [29]. The certified value's *denominator* is bounded through L_{asm} together with the height of the placed coordinates, but its *bit-size* additionally needs the horizon: an admissible closing family holds $L_{\mathrm{asm}} = 2$ and $\mathrm{ht}(x^\star) = 1$ at every horizon while $t^\star = N$ exactly. Nor is either quantity bounded through L alone: integer goal weights of growing magnitude take $\mathrm{den}(t^\star)$ unbounded at fixed $L = 1$, since goal recovery divides by W_k and the weight's numerator lands in the value's denominator.

No unconditional polynomial bit bound exists: quadratic optimization *at large* can force 2^n-bit exact solutions [6], and small-rational-solution existence is strongly NP-hard. What we *can* support is Theorem 1's guarantee for the constructed families, and the reason it is supportable is the design order of Definition 3: $(\lambda^\star, x^\star)$ is placed at per-stage height $O(L)$ before any datum is derived from it, so no solve stands between the design tuple and the emitted pair.

Appendix B Weak diagonal dominance: a checkable entry to A1

Role in this paper. Aggregate-PSD (A1) is the closure theorem's key hypothesis, and a conditional theorem is usable only with a checkable hypothesis. Weak diagonal dominance is that entry: solver-free, proved, preserved along the sweep. It is an architectural device, not a frequency claim — how often real instances satisfy it is unmeasured, and the demonstration of §4 verifies A1 by the general sweep rather than by this shortcut. We call an instance *benign* when aggregate-PSD (A1) holds at its optimum — the S-procedure is then exact and the bracket can close. This is a property of the instance, not of a solver; this section gives a sufficient, solver-free condition checkable stage-by-stage along the sweep.

Lemma 2 (Solver-free benign criterion)**.** *Call H* weakly diagonally dominant *(WDD) when $H_{ii} \geq \sum_{j \neq i} |H_{ij}|$ on the* scalar *entries of the assembled H, not its $d \times d$ blocks; the left side is H_{ii}, not $|H_{ii}|$, so the inequality already forces a nonnegative diagonal. If H is symmetric and WDD, then $H \succeq 0$, and the Schur complement after each scalar pivot of the sweep is again WDD, so every pivot along the sweep is nonnegative.*

For the identity-coupled chain ($A_k = I$, $R_k = I$) with full-stage keep-outs ($\Pi_j = I$), WDD is exactly the per-stage condition $w_k \geq \lambda_k$, where w_k is the scalar goal weight ($W_k = w_k I$) and λ_k the total multiplier load on stage k: the goal weight dominates the keep-out multipliers. Under a projection Π_j the load falls only on the guarded coordinates, so WDD becomes the per-coordinate condition $w_k \geq \max_i \lambda_{k,i}$, with $\lambda_{k,i}$ the summed multipliers of the keep-outs guarding coordinate i of stage k. The stage condition $w_k \geq \lambda_k$ remains sufficient, and remains sharp whenever some coordinate is guarded by every keep-out on the stage — in particular for a single projected keep-out; it is strictly conservative exactly when the guarded sets of the keep-outs carrying positive *multipliers share no coordinate, which, those sets being nonempty, forces at least two positive multipliers.*

Proof. A symmetric WDD matrix is PSD by Gershgorin (machine-checked; [29]). The Schur complement of a strictly diagonally dominant matrix is strictly diagonally dominant [7]; the weak case follows by perturbing the diagonal by $\varepsilon \to 0^+$, zero pivots having zero rows in the symmetric WDD case and being skipped, as in Lemma 1. Applying this one pivot at a time keeps every Schur complement WDD, hence every pivot nonnegative; $H \succeq 0$ itself is the Gershgorin step above. In the identity-coupled scalar specialization the whole check reduces to the local test $w_k \geq \lambda_k$ on each row of H — interior rows $[-1,\, 2 + w_k - \lambda_k,\, -1]$ and the two boundary rows $[\,1 + w_k - \lambda_k,\, -1\,]$ — with no block pivot S_k computed, which also avoids presuming the nonsingularity that S_k's recursion would need. □

WDD is sufficient, not necessary. The one case where it is also necessary is special: for the identity-coupled free-endpoint chain with *uniform* excess $w_k - \lambda_k \equiv c$, the path-Laplacian critical mode is the constant vector, so $H \succeq 0$ iff $c \geq 0$. That is a boundary statement, and its hypotheses — identity coupling, uniform excess — are load-bearing.

One keep-out per stage does not guarantee A1 either; the next example certifies a duality gap on exactly that geometry.

Example 1 (Certified duality gap)**.** A $d{=}1$, $N{=}2$ instance of (1) with one keep-out per stage, stage-coupling coefficient $-3/2$, and nonzero goals has $J^\star = 212/13$, by exact enumeration of the branch-KKT candidates. A rational moment point caps *every* level-one certificate at $815/64$ by weak duality. The level-one gap is thus certified $\geq 2973/832 \approx 3.57$, entirely over $\mathbb{Q}$.

The same receipt discharges the step from a gap to a failure of A1 rather than leaving it to the reader: $x^\star$ is rational (A2) and exactly one keep-out is active there, so LICQ is automatic and Theorem 1 would close the bracket if A1 held. It does not: stationarity recovers $\lambda^\star = 24/13$ and $H(\lambda^\star)$ has determinant $-7/4$, so $H(\lambda^\star) \not\succeq 0$. This is why aggregate-PSD (A1) is a real assumption and not automatic: the block-banded smoothness coupling is not itself a gap source under WDD, but general coupling with nonzero goals can break it.

Hard dynamics. Hard equality dynamics $x_{k+1} = A_k x_k + B_k u_k$ are outside (1) by definition: they add control variables to a decision vector §2 takes to be the stage states alone, and an equality to a class that carries only inequalities. Lemma 2's criterion is likewise stated for the penalty-coupled assembly only. The *closure* argument is less fragile than that: an affine equality contributes nothing to the Lagrangian curvature matrix and enters weak duality with a free multiplier, so wherever the analogue of aggregate-PSD holds at a rational KKT point the same constant-multiplier bracket closes. An exact receipt certifies both halves over $\mathbb{Q}$ on $d{=}1$ instances with one keep-out and one equality [29]. On the first, the best constant-multiplier bound is exactly 1 while $J^\star = 3/2$, a certified gap of $1/2$. The diagnosis, though, is Example 1's, not a statement about hard coupling: stationarity there recovers $\lambda^\star = 3/2$ against a unit curvature, so $H(\lambda^\star) = \mathrm{diag}(-1/2, 1, 1)$ is not PSD and A1 fails, with A2 and LICQ intact. On its sibling — the same variables and the same equality, with the goal moved and the keep-out radius doubled — stationarity recovers $\lambda^\star = 3/4$ and $H(\lambda^\star) = \mathrm{diag}(1/4, 1, 1) \succeq 0$, so A1 holds and the constant-multiplier bracket closes exactly at $t^\star = J^\star = 9/4$: the same trusted dense M-test accepts $t^\star$ and refuses $t^\star + 10^{-6}$. The exhibited gap therefore measures A1, as Example 1's does, and no conclusion about Theorem 1 follows from it.

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